Cognitive Disorder Children's Rehabilitation Training Answering System

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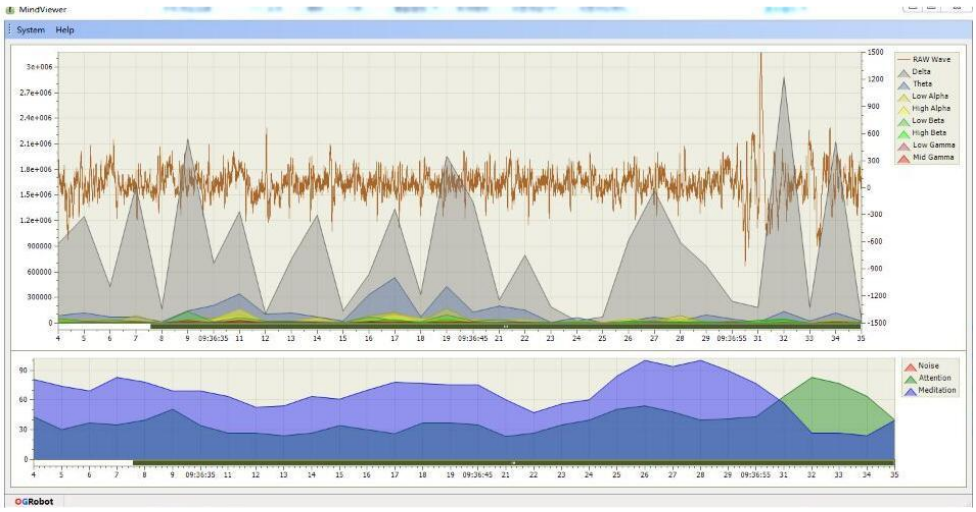
This semester, my professional course assignment is also a long - term project selected for the "Music Talent Flying Plan", the "Cognitive Disorder Children's Rehabilitation Training Answering System", which aims to break through the limitations of existing rehabilitation training models with emerging technologies.

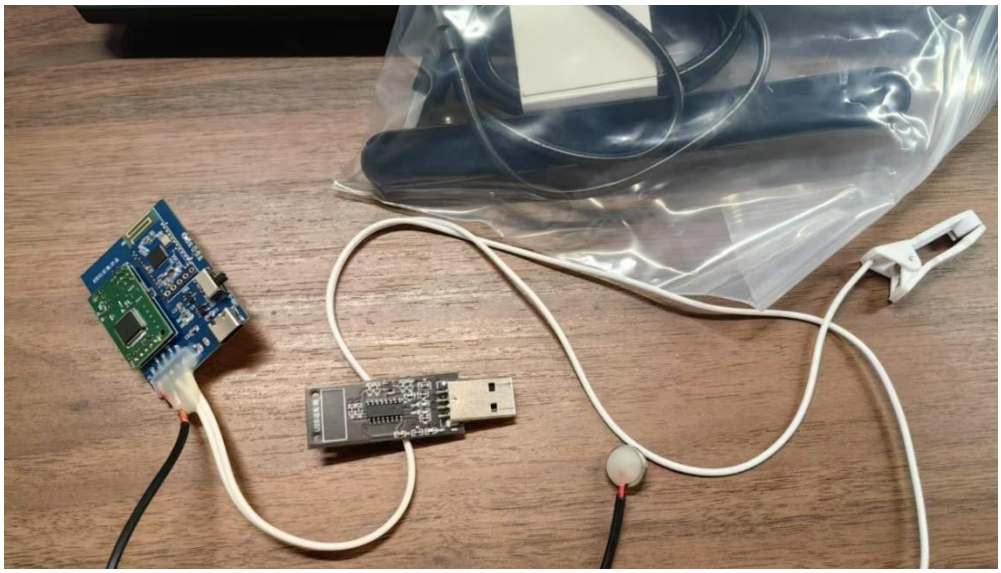
The project originated in November 2023. I conducted an in - depth investigation of the Rehabilitation Department of Shanghai Children's Hospital with my supervisor. I noticed that traditional answering systems for cerebral palsy children's rehabilitation training mainly rely on visual stimuli. For example, children are shown fruit pictures and required to select the corresponding fruit sections. After consulting a large number of literatures and communicating with the hospital, I determined and proposed that introducing multi - sensory synesthesia into this system could enhance the training effect.

In recent years, the state has continuously increased its support for medical rehabilitation technology innovation. However, currently, the rehabilitation training for children with cognitive disorders often lacks personalized and precise plans, resulting in uneven rehabilitation effects. Starting from this point, I first began to research and develop a rehabilitation training system for cerebral palsy children. The new system I designed added diversified options of game mode and training mode, and changed the question types. Children will select the corresponding patterns from visual options according to the sounds they hear. This small change aims to accelerate neural remodeling and improve information processing and cognitive abilities through the coordinated action of multiple senses. In November, I brought the above - mentioned program demo to Shanghai Children's Hospital again. Under the guidance of doctors' suggestions, the project policy was changed to "Cognitive Disorder Children's Rehabilitation Therapy", and the feasibility and effectiveness of this answering system were determined, and further production and improvement began.

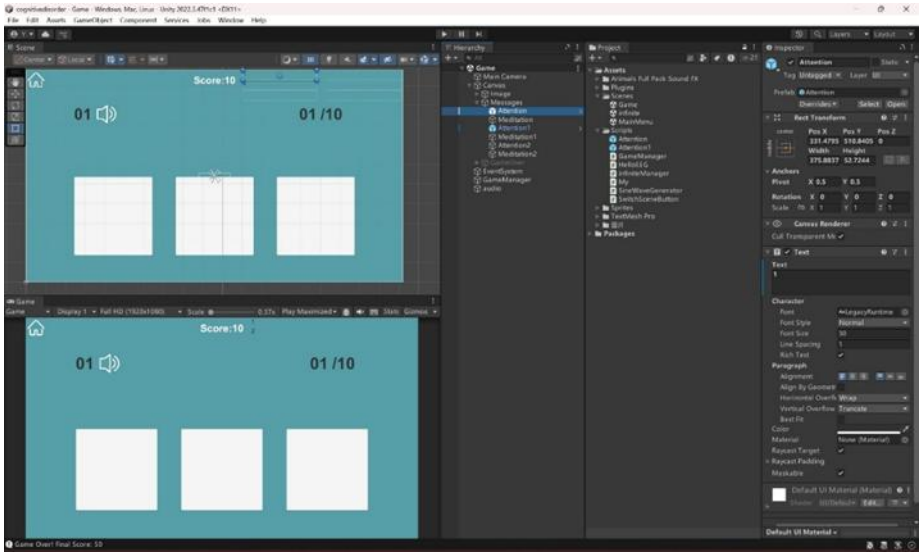


During this period, under the guidance of Tianqiao Institute of Brain Science and East China Normal University, I incorporated brain - computer interface (BCI) technology into the project. BCI technology

enables us to detect children's brainwaves in real - time, providing reliable data support for formulating personalized rehabilitation training. Various research data also confirm its help in rehabilitation engineering. Therefore, we envisioned the application of BCI in the rehabilitation training of cerebral palsy children. It can not only help children control robotic arms for motor rehabilitation but also achieve non - invasive thinking communication. The completed answering training system can also access algorithms through the BCI. By analyzing real - time brain - wave data, the questions can be optimized in real - time, making the training more targeted and effective.



In terms of program production, I downloaded the answering system framework from GitHub. I modified the visual text questions into automatically playing audio, the text options into pictures, added a scoring system, and adjusted the appearance and the number of options.

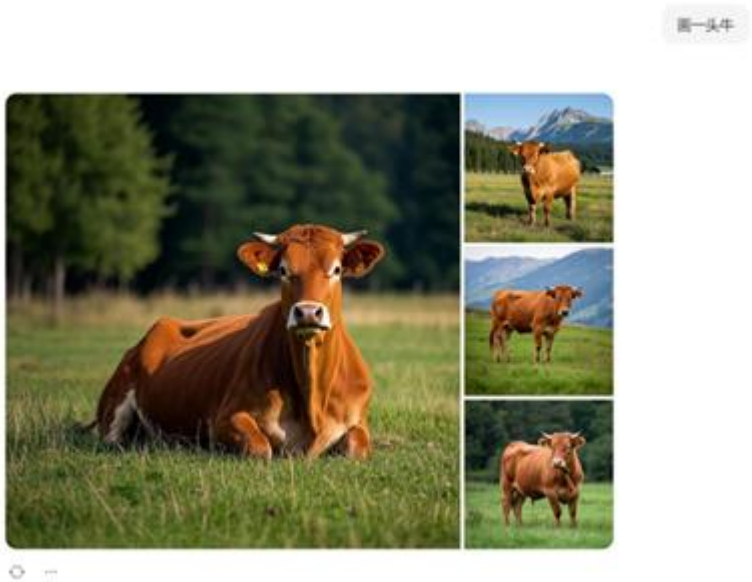


The driver of the BCI was a bit challenging. Since the device was purchased from a second - hand platform, some parts were damaged, and there was no instruction manual. I couldn't normally import the brain - wave data into Unity. Until late November, I sought help from a professor at East China Normal University and obtained a re - written C# driver script, which allowed the BCI device to operate successfully. After that, I learned to write scripts for importing and displaying data, and successfully displayed two brain - wave data in the game for reference by rehabilitation doctors, accompanying personnel, and for further use in the future. In the 1.0 version demonstrated this semester, I set the focus level multiplied by 20 as the frequency and the relaxation level minus 70 as the amplitude, and added a sine wave that changes in real - time as an example of the plasticity of brain - wave data.

According to the smart EEG BCI device, specific sound wavelengths can help achieve deep sleep. Thus, I reasonably hypothesized that specific sine waves can also stimulate brain activity to help children concentrate. The problem of children's non - cooperation in current rehabilitation training has not been solved. If

this hypothesis holds true, it will bring great benefits. This hypothesis still needs to be further confirmed through communication with the hospital. On the other hand, I will expand the question bank in the future and use these two data as the criteria for changing question types.

Most of my sound materials are sourced from the Internet. The sounds of cows and ducks were recorded by me with a microphone in Inner Mongolia in August last year. The animal pictures are generated by AI.



My final answering program is divided into a game mode and a training mode. The game mode has ten questions and calculates the score at the end. The training mode imitates the infinite - form of the original hospital system. The code and framework have been completed, but due to the lack of a question bank, it could not be fully presented this semester.

In the future, I will improve this project in three aspects:

1. Expand the question bank and add more question types. With the help of Tianqiao Institute of Brain Science, I will try to introduce the data collected by the BCI into the algorithm to change the question types in real - time and create a customized rehabilitation experience for children. For example, if a child shows higher focus and relaxation levels in questions about animated characters, we will reduce the number of other types of questions, such as animal - related questions.
2. Improve the training mode after the question bank meets the standard. I learned from my visit to the hospital that the game mode is more suitable for children to play at any time with their parents' accompaniment, while the training mode is more suitable for the hospital's rehabilitation department to directly obtain data and evaluate the training effect.
3. Try to incorporate the use of an eye - tracker. When I bought the BCI, I also purchased an eye - tracking device. Since the existing system still requires children to operate with their fingers, it is quite difficult for some children who have other disabilities in addition to cognitive impairments. The introduction of eye - tracking can provide more training forms and increase attention and fun.

I hope to break through the bottleneck of cognitive function rehabilitation for cerebral palsy children with BCI and eye - tracking technologies. In the future, I will further cooperate with Shanghai Children's Hospital and Tianqiao Institute of Brain Science, striving to apply this innovative achievement to clinical practice as soon as possible.